Proposed Requirements Document: Emergency Ambulance Response System (EARS)

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1 Project Overview

Project	Emergency Ambulance Response System (EARS)
Modules	Patient App, Driver App, Admin/Dispatcher Panel
Author	
	Position: Project Lead, M Holdings **Contact**: info@mholdings.co.za

2 Purpose

This proposal aims to transform emergency medical services by developing the Emergency Ambulance Response System (EARS), a real-time, cross-platform solution to enhance ambulance request and dispatch processes in Thohoyandou and beyond. EARS aligns with the strategic goal of improving healthcare access and response times through digitalization, addressing inefficiencies in manual emergency response systems.

3 Objectives

The primary objective of EARS is to streamline emergency medical services by automating ambulance requests, tracking, and dispatch processes. Specific goals include:

- Developing a cross-platform tool (Android, iOS, Web) for the public, ambulance drivers, dispatchers, and health institutions.
- Enabling secure storage and transmission of patient medical data.
- Reducing response times through real-time tracking and optimized dispatch.
- Ensuring accessibility with multilingual support (e.g., Tshivenda) and offline capabilities.
- Complying with data privacy regulations, such as POPIA.

4 Definitions

- **EARS** - Emergency Ambulance Response System
- **FIN** - Financial Department (for potential funding integrations)
- **ICT** - Information and Communication Technology
- **POPIA** - Protection of Personal Information Act
- **HI** - Health Institutions (hospitals, clinics)

5 Background

The need for EARS stems from the strategic objective to digitize and improve emergency healthcare services in South Africa, particularly in underserved areas like Thohoyandou. Current manual processes for ambulance requests lead to delays, lack of transparency, and

inefficiencies. Through stakeholder consultations with community members, healthcare providers, and emergency responders, the following pain points were identified:

- Slow response times due to manual dispatch processes.
- Lack of centralized patient medical data for responders.
- Limited accessibility for rural and vulnerable populations.
- Inadequate tracking of ambulance locations and statuses.
- Challenges in addressing mental health emergencies discreetly.
- Need for multilingual and offline support in remote areas.

The ICT division, in collaboration with M Holdings, proposes EARS to address these challenges through a robust, automated system. The system ownership lies with M Holdings, with ICT providing technical support for development, analysis, and implementation.

6 User Story

- **Patient Perspective:** As a member of the public, I want to access a user-friendly app to request an ambulance (private or government) in emergencies, so that my location and medical history are automatically sent to responders for quick and informed assistance.
- **Driver Perspective:** As an ambulance driver, I want a tablet-based app to receive nearby emergency requests with patient details and navigation, so I can respond efficiently and update request statuses.
- **Dispatcher Perspective:** As a medical dispatcher, I want a web-based panel to monitor ambulance locations, assign drivers, and review requests, so I can manage operations effectively.
- **Health Institution Perspective:** As a hospital, I want to receive patient data from EARS to prepare for incoming emergencies.

Users access EARS through a secure login portal, select their role (patient, driver, dispatcher), and interact with role-specific modules. Patients can trigger emergency requests, drivers can accept and navigate, and dispatchers can oversee operations.

7 Functionality Requirements

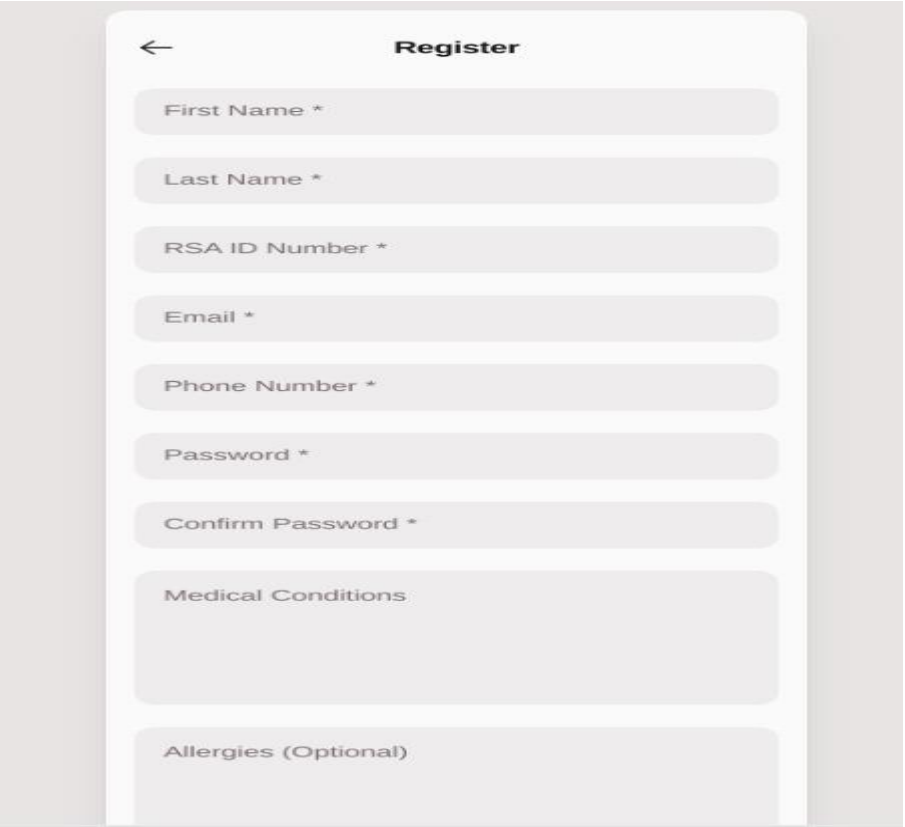
7.1 Process Mapping for EARS

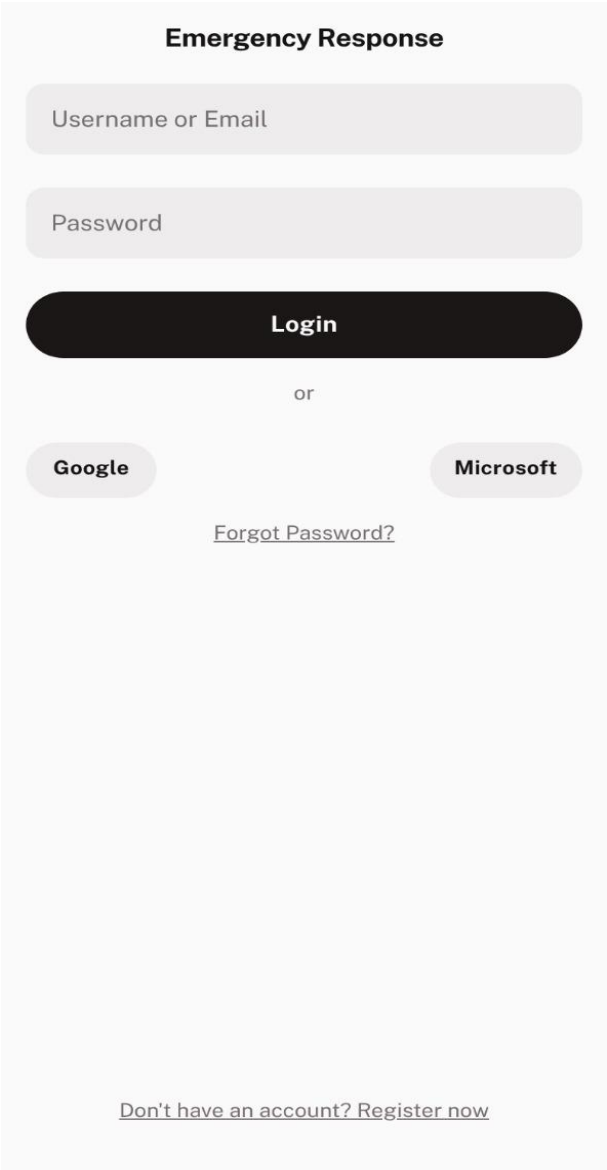
- **Patient Flow:** Register → Select ambulance type → Update medical profile → Trigger emergency request → Receive real-time updates.
- **Driver Flow:** Login → View nearby requests → Accept/reject request → Navigate to patient → Update pickup/drop-off status.
- **Dispatcher Flow:** Login to web panel → Monitor ambulance locations → Assign drivers → Review request history.

7.2 Business Rules

- Users must be registered with valid credentials to access EARS.
- Patient medical data is encrypted and only shared with authorized responders.
- Mandatory fields (denoted by *) must be completed for registration and requests.
- Anonymous requests are allowed for mental health emergencies to ensure privacy.
- Drivers can only accept requests within a defined proximity.

7.3 EARS Interfaces



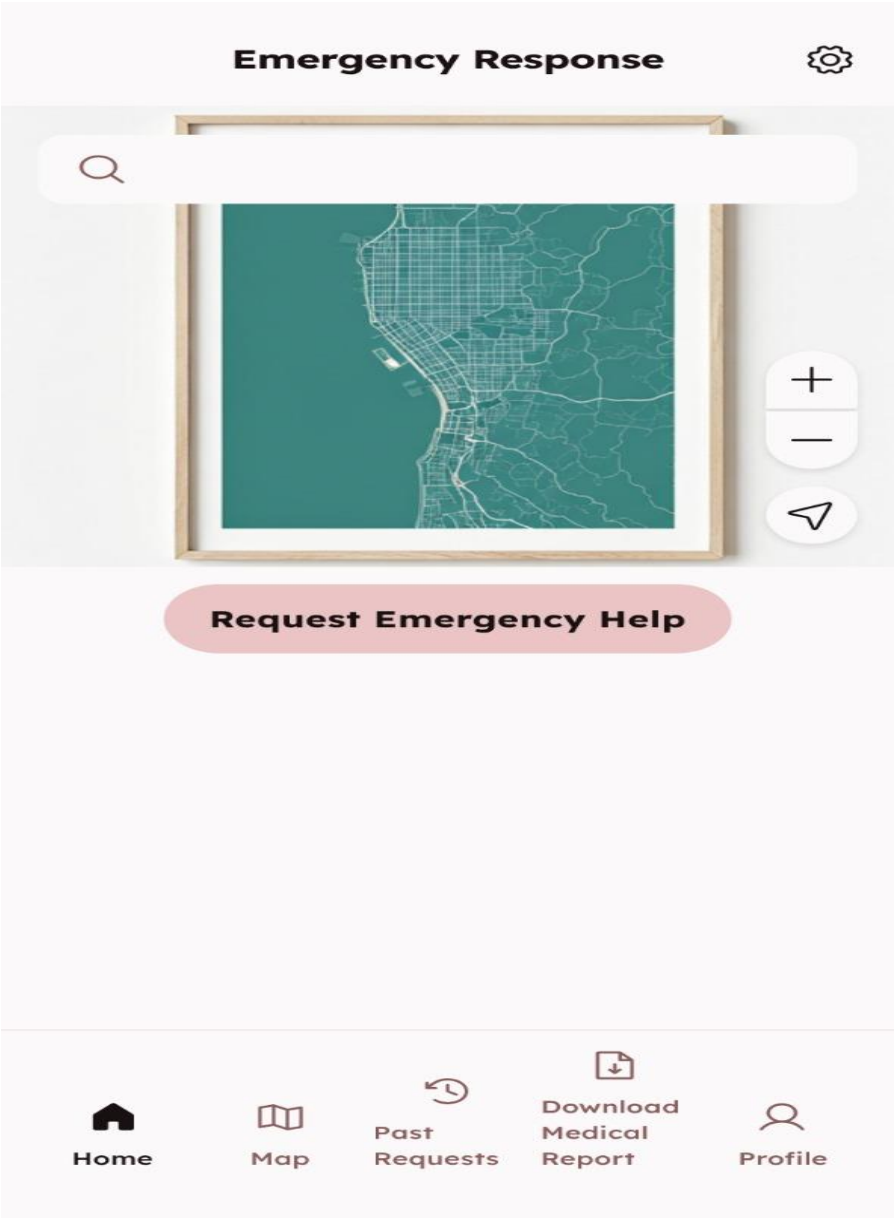


7.3.1 Login Interface

- Secure login for patients, drivers, and dispatchers using Firebase Auth or OAuth2.
- User profile auto-loaded upon successful login.

7.3.2 Database for User Login

Field Name	Data Type	Description
user$_id$	INT (PK, Auto)	Unique identifier for the user
username	VARCHAR(50)	Login username
password	VARCHAR(255)	Hashed user password
role	ENUM	Role (Patient, Driver, Dispatcher, HI)
created$_at$	TIMESTAMP	Timestamp when user was added

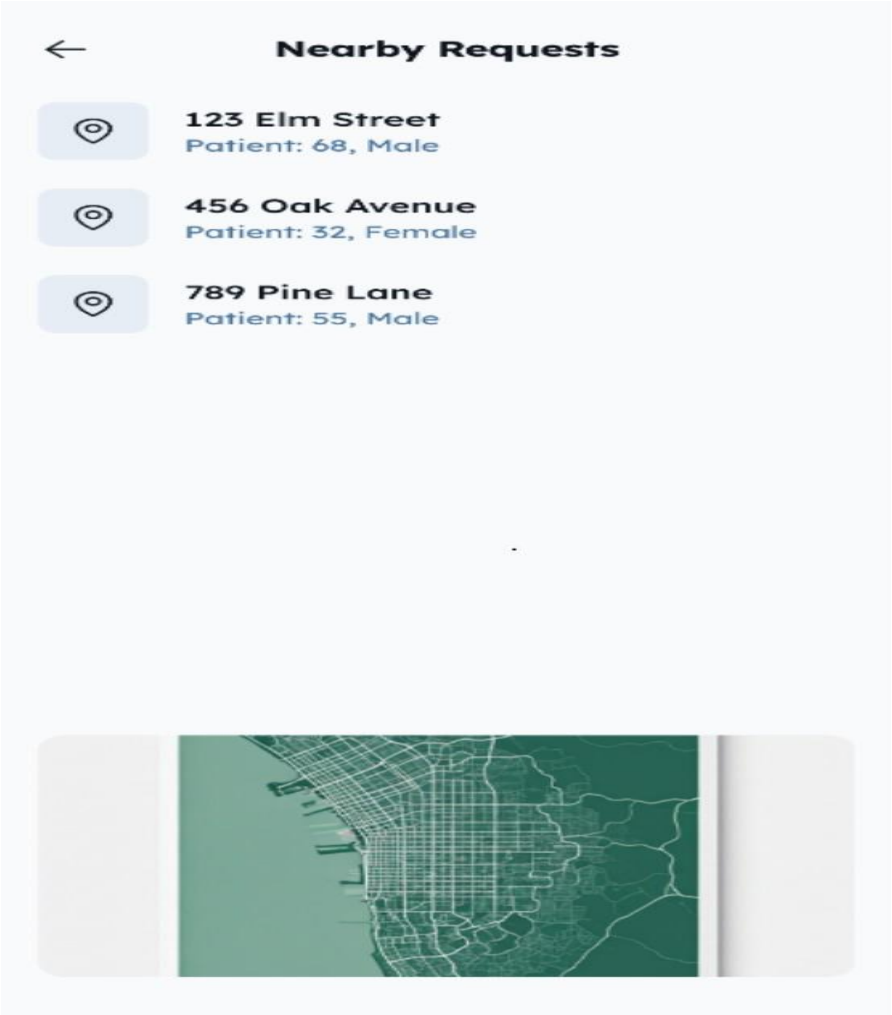


7.3.3 Patient App Interface

- **Home Page:** Displays a large, red "EMERGENCY" button, profile update option, and request history.
- **Emergency Request:** Auto-populates user details and medical history; sends location and data to the nearest ambulance.
- **Notifications:** Real-time updates on ambulance status and ETA.

7.3.4 Patient Requests Table

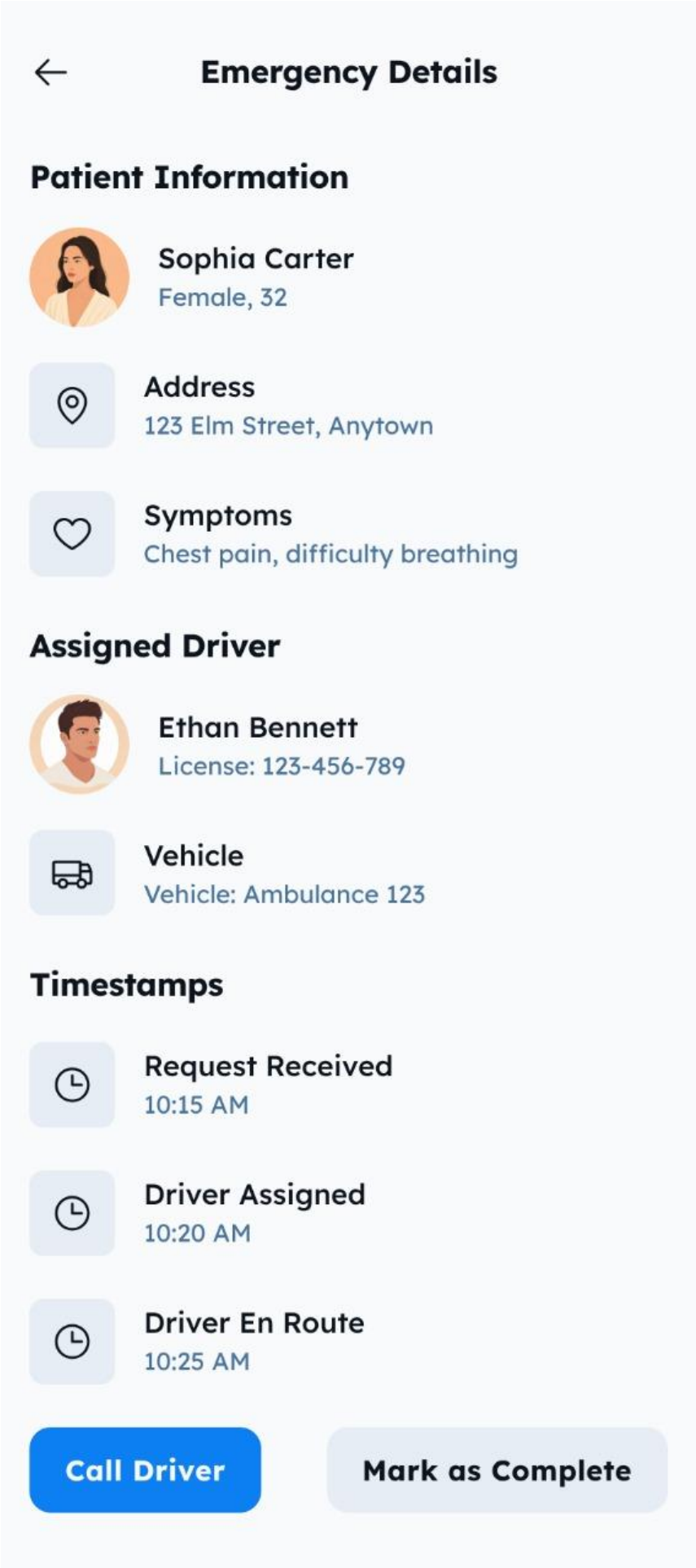
Field Name	Data Type	Description
request_id	INT (PK, Auto)	Unique identifier for the request
user_id	INT (FK)	Patient who made the request
ambulance_type	ENUM	Ambulance type (Private, Government)
medical_data	TEXT	Patient medical history (allergies, conditions)
location	VARCHAR(255)	Patient's current location
status	ENUM	Request status (Pending, Accepted, En Route, Completed)
created_at	TIMESTAMP	Timestamp of request submission
driver_id	INT (FK)	Assigned driver (if accepted)
accepted_at	TIMESTAMP	Acceptance timestamp



7.3.5 Driver App Interface

- Displays nearby emergency requests with patient details.
- Integrated GPS navigation to patient location.

- Options to accept/reject requests and mark pickup/drop-off.


Emergency Details
Patient Information
Sophia Carter
Female, 32
Address
123 Elm Street, Anytown
Symptoms
Chest pain, difficulty breathing
Assigned Driver
Ethan Bennett
License: 123-456-789
Vehicle
Vehicle: Ambulance 123
Timestamps
Request Received
10:15 AM
Driver Assigned
10:20 AM
Driver En Route
10:25 AM
Call Driver
Mark as Complete

7.3.6 Dispatcher Panel Interface

- Real-time map view of ambulances and active requests.
- Options to assign drivers, review requests, and manage user accounts.

7.3.7 Ambulance Assignment Table

Field Name	Data Type	Description
assignment$_id$	INT (PK, Auto)	Unique identifier for assignment
request$_id$	INT (FK)	Related request ID
driver$_id$	INT (FK)	Assigned driver
assigned$_by$	INT (FK)	Dispatcher who assigned
assigned$_at$	TIMESTAMP	Assignment timestamp

8 Integration

- **Hospital Systems:** Integrate with health institution ERPs via REST APIs to share patient data.
- **Location Services:** Use Google Maps API for real-time tracking and navigation.
- **Authentication:** Leverage Firebase Auth or OAuth2 for secure login.
- **Push Notifications:** Implement Firebase Cloud Messaging for real-time updates.

9 Recommendation

We request stakeholders, including M Holdings and potential health institution partners, to review this requirements document and provide feedback. Upon approval, we seek authorization to proceed with the development lifecycle, including the Thohoyandou pilot.

10 Conclusion

EARS is a transformative solution to enhance emergency medical services in South Africa. By addressing stakeholder pain points and leveraging modern technology, EARS will improve response times, accessibility, and transparency. We urge stakeholders to approve this document to initiate development and meet strategic healthcare objectives.

Date: *July* 2025